Tipping an undecided swarm

Simulation 1, setup `task003-nosolution`: first results

LLM-free simulator; extended with the efficiency of control, 12 October 2026; runs of 7 October 2026

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact about it and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. An episode has 30 rounds, each a day and a night. During the day the agents act one at a time in a new random order: an agent reads q posts drawn at random from a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* (*argmax*: the most probable allocation; *probability matching*: a vote drawn from its belief) and posts one remembered fact that supports its vote, or nothing if none does. At night a *controller* with a target, A0 (truth) or A2 (false), reads q_c agent posts; unless at least 75% of them come from agents voting for its target, or what it read already proves its target, it posts the b facts from its own pool that best support its target. The board is then cleared, keeping only the controller's new posts. At each dawn every remembered fact is forgotten with probability $1 - \rho$ and comes back if read again. Every episode is run three times with the same random draws: without a controller, with the truth controller and with the false controller.

Summary. 24 agents hold evidence that, pooled, is exactly a coin flip between allocation A0 (the truth) and A2, with no proof available to anyone. A controller steers toward A0 or A2 using one shared pool of facts. With agents that vote by **probability matching**, the swarm stays undecided on its own, and **either controller tips it within a few days**; the truth controller does so more easily. With *argmax* voting and vote-supporting posts, the swarm instead locks into unanimity by itself, and a design detail (an expected start of 7 / 6 / 11 votes) makes it drift to A2; we explain this special case and how to remove it. Two further results come from the design rather than the swarm: **a larger posting budget can backfire** because the pool has few facts per target; and the controller **senses the swarm accurately**.

1 Setup

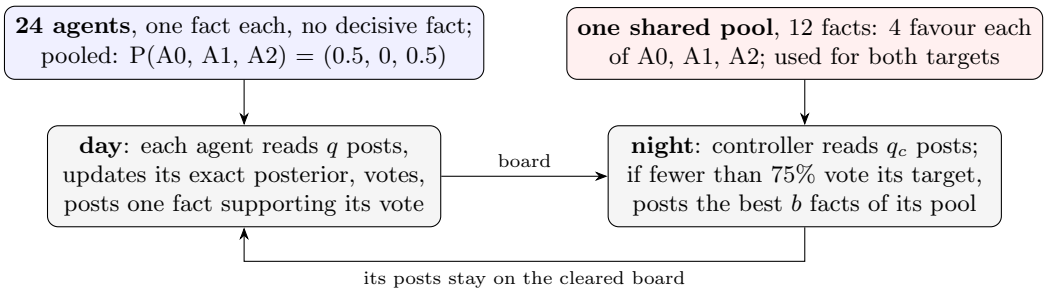


Diagram 1. One round of Simulation 1, repeated for 30 rounds. Each dawn, every remembered fact survives with probability ρ.

No proof is available to anyone, not even if the controller posted its whole pool: the agents' facts plus the whole pool give (0.75, 0, 0.25). The truth keeps an edge that cannot be designed away (every fact is true): pooled agents plus the best part of the pool reach at most P(A0) = 0.92, but P(A2) = 0.77.

Agents are exact Bayesian reasoners with posterior $P_j = (P_j(\mathrm{A0}), P_j(\mathrm{A1}), P_j(\mathrm{A2}))$ over the facts they remember. They vote by one of two rules:

$$\text{probability matching:} \quad \Pr(\text{vote}_j = k) = P_j(k), \tag{1}$$

$$\text{argmax:} \quad \text{vote}_j = \arg\max_k P_j(k), \text{ ties broken at random.} \tag{2}$$

Control is measured by the *paired gain*: each episode is run silent, with the truth controller, and with the false controller, using the same random draws, and

$$G_t^{(a)} = \frac{1}{E} \sum_{e=1}^{E} \Big[x_{a,t}^{\mathrm{ctrl}}(e) - x_{a,t}^{\mathrm{silent}}(e) \Big], \tag{3}$$

where $x_{a,t}$ is the share of agents voting a on day t, $E = 1000$ episodes.

2 Main result: with probability matching, either controller tips the swarm

Without a controller the swarm stays undecided. Under probability matching no episode ever becomes unanimous, and the A2 − A0 vote margin stays near zero (Figure 4, red): exactly 0 with full memory, and a small lean of +0.07 to +0.15 toward A2 when agents forget (A2-side facts survive forgetting better: held by 53% of agents on day 30, A0-side facts by 31%).

With a controller it tips within days. Figures 1 and 2 show the day-30 gain for every reading budget q_c and posting budget b; Figure 3 shows how the gain builds up over the days.

Table 1: Probability matching, day 30, $q_c = 12$, $b = 3$: paired gain in the target's share, Eq. (3), for $q = 3/6/12$; and the share of episodes in which the target passes 75% by day 30 (mean over q), with the silent value in brackets.

ρ	controller	gain ($q = 3/6/12$)	target past 75%	median day
0.75	truth (A0)	0.33 / 0.42 / 0.43	0.99 (0.26)	3–4
0.75	false (A2)	0.21 / 0.25 / 0.23	0.98 (0.58)	3
1.0	truth (A0)	0.37 / 0.37 / 0.38	1.00 (0.36)	3
1.0	false (A2)	0.08 / 0.07 / 0.06	0.90 (0.47)	3

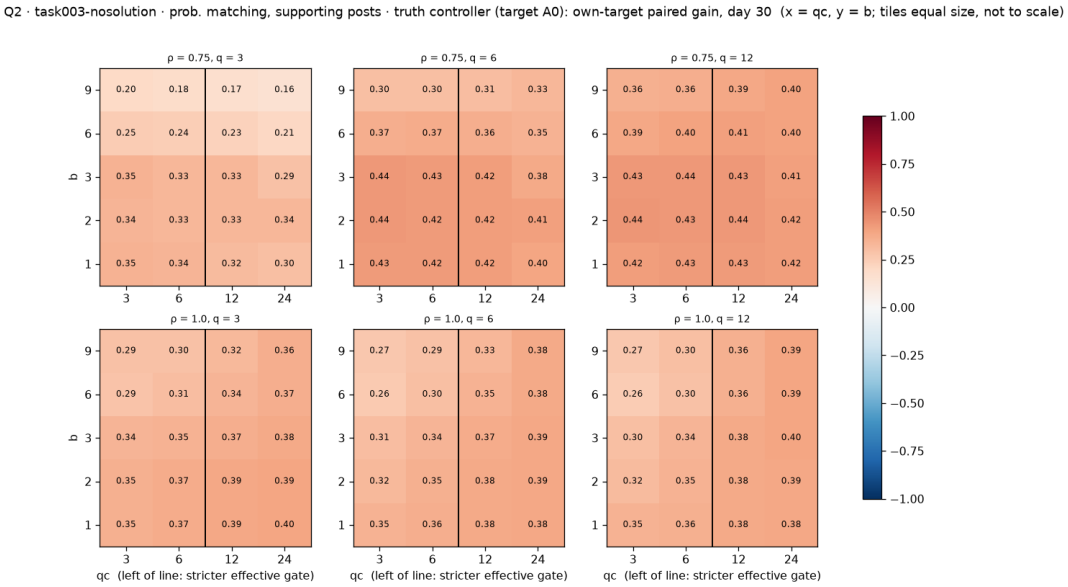


Figure 1: Truth controller, probability matching: day-30 paired gain in the A0 share over q_c (x) and b (y). Rows: ρ; columns: q. Tiles equal size, not to scale. Hatched: 95% interval includes 0. Left of the black line the vote gate is effectively stricter.

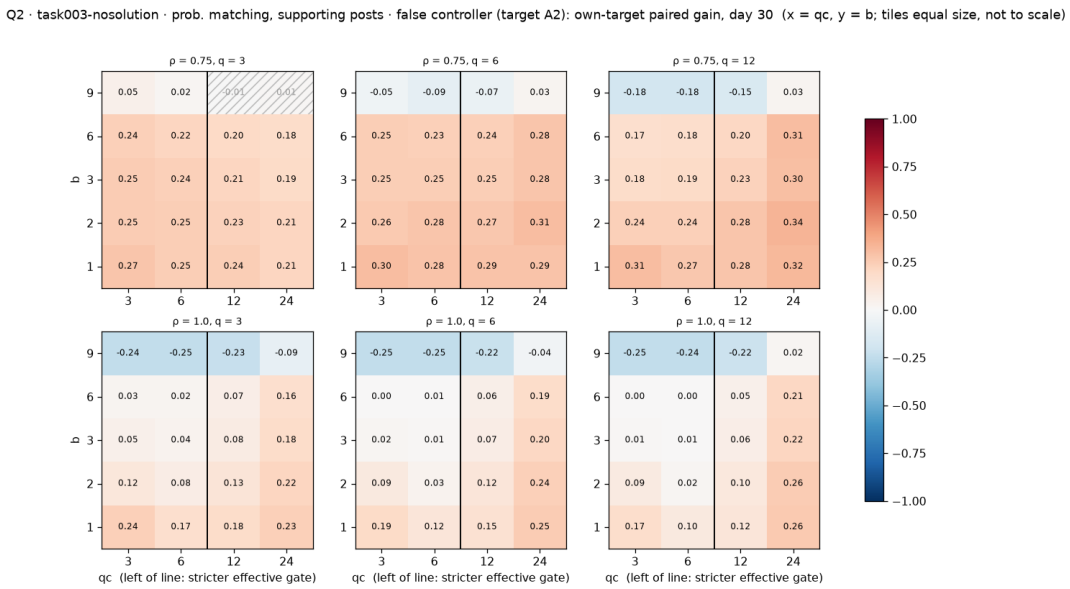


Figure 2: As Figure 1, for the false controller (gain in the A2 share).

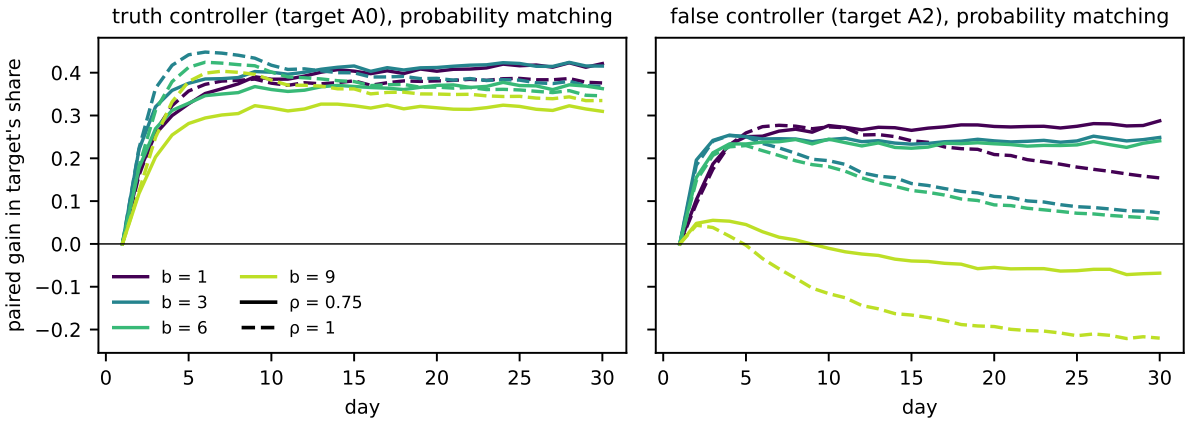


Figure 3: Probability matching: paired gain in the target's share over the days, by posting budget b; $q = 6$, $q_c = 12$. Solid $\rho = 0.75$, dashed $\rho = 1$.

- **Both controllers tip the swarm in about 3 days** (Table 1).
- **The truth controller is stronger**, and the gap widens with full memory: gains of about 0.37 against 0.07 at $\rho = 1$. As agents accumulate facts, their pooled evidence pulls back toward the truth's higher ceiling; the false controller's gain decays over the days (Figure 3, right, dashed).
- **The reading budget q_c matters little**; the posting budget matters only at its largest value (Section 4).

3 A special case: the A2 drift under argmax voting

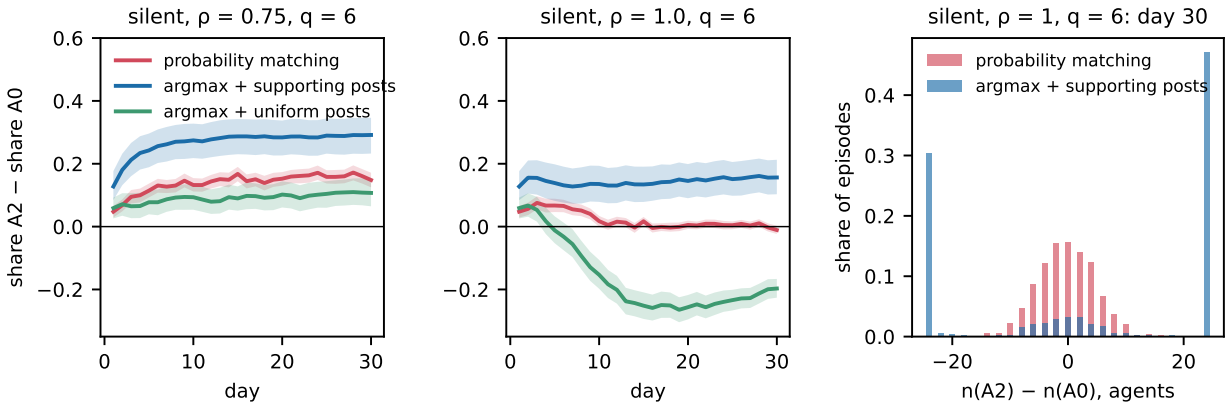


Figure 4: Silent arm, $q = 6$. Left and middle: A2 − A0 vote margin over the days. Right: day-30 spread of the margin across episodes, in agents.

With argmax voting and vote-supporting posts (the *base* rule), the silent swarm does not stay undecided: **63–91% of episodes end unanimous**, all 24 agents on A2 (43–67%) or on A0 (15–41%). The mean margin toward A2 (Figure 4, blue) is the difference between those two groups; the histogram shows the lock-in at ±24. Three ingredients combine:

1. **Argmax turns any lead into a vote**, Eq. (2).
2. **Agents post only facts supporting their vote**, so later readers mostly meet the majority's evidence, and with forgetting the minority's evidence dies out (day 30, $\rho = 0.75$: A2-side facts held by 53% of agents, A0-side by 24%).
3. **The swarm starts with more A2 voters.** Under argmax, an agent holding a fact that favours two allocations equally votes for each half the time. Its expected starting vote is

$$\mathbb{E}[n_k] = \sum_{j=1}^{24} \frac{\mathbb{1}\{k \in \arg\max P_j\}}{|\arg\max P_j|}, \tag{4}$$

   which gives **7 / 6 / 11** for A0 / A1 / A2 (Table 2).

Remove ingredient 1 (probability matching) or 2 (uniform posting: post any remembered fact) and the lock-in collapses: no unanimous episode at all under probability matching, 9–22% under uniform posting.

Table 2: Why the start is 7 / 6 / 11. The setup balanced agents by *fact group* (8 per allocation), assigning a tied fact to the lower-numbered allocation; argmax votes split ties.

	A0	A1	A2
agents per fact group (design bookkeeping)	8	8	8
of which hold a fact tied A1/A2 (counted as A1)	–	6	–
of which hold a fact tied A0/A1 (counted as A0)	2	–	–
expected starting votes, argmax, Eq. (4)	**7**	**6**	**11**
expected starting votes, probability matching	8.3	7.4	8.3

How to fix the start. Balance the agents on *expected votes*, Eq. (4), instead of on fact groups. Under the current rules this is impossible: none of the 10,438 coin-flip agent sets the builder can produce has equal expected A0 and A2 votes. The reason is scarcity: only 3 eligible facts favour A1 alone, so any 5-fact A1 group must include A1/A2-tied facts. Dropping the fact-group rule and requiring expected votes of exactly 8 / 8 / 8 directly works: a random search found such agent sets (still a coin flip, all facts eligible), e.g. 12 facts each held by 2 agents. A full rebuild would also re-check the pool rules (no shared facts, no joint proof). This changes the frozen setup, so it is a decision, not yet applied. Probability matching needs no fix: its start is already balanced.

4 More budget can be worse: a design effect

At the largest posting budget the controller's gain falls (Figures 1, 2, top row; Figure 3, $b = 9$), and the false controller's can even turn negative (-0.08 at $\rho = 0.75$, -0.22 at $\rho = 1$, $q_c = 12$, mean over q). The cause is arithmetic, not swarm dynamics. The controller must post exactly b facts or none, and the shared pool holds only n_{target} facts on its target's side, so each post contains at least

$$c = \max(0,\ b - n_{\text{target}}) \tag{5}$$

facts that do not favour its target.

Table 3: Pool composition and what the false controller actually posts (probability matching, $\rho = 1$, $q = 6$, $q_c = 12$). Tied facts count for both sides.

	$b = 3$	$b = 6$	$b = 9$
facts in the pool on the target's side (n_{target})		A0: 4, A2: 6	
posted facts favouring A0 / A1 / A2	14 / 23 / 62%	12 / 22 / 67%	**26** / 23 / 51%
nights the controller posts	59%	63%	**93%**

The fall starts exactly where Eq. (5) becomes positive: at $b = 6$ for the truth controller ($n = 4$), at $b = 9$ for the false one ($n = 6$). Its counter-evidence keeps the target's vote share below the gate, so it keeps posting (Table 3), and the counter-evidence accumulates. The effect replicates with fresh random draws (a second seed). It disappears if the controller may post *up to* b facts, or never posts facts against its target; in task003-symmetric, whose pools are entirely on their target's side, more budget simply helps.

5 The controller senses the swarm accurately

Each night the controller estimates the share of agents backing its target from the $n \le q_c$ agent posts it reads, $\hat{v} = \frac{1}{n}\sum_i \mathbb{1}\{\text{author of post } i \text{ voted for the target}\}$, and stays silent when $\hat{v} \ge 0.75$.

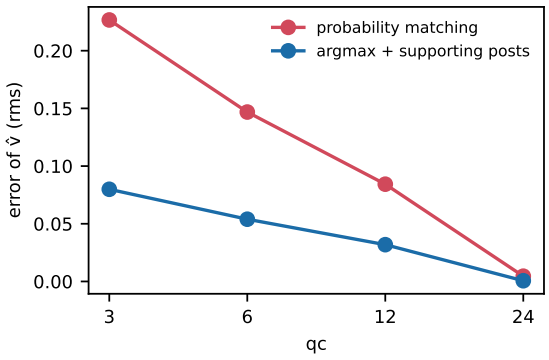


Figure 5: Error of $\hat{v}$ against the true share of all 24 agents, task003-nosolution.

$\hat{v}$ is unbiased (± 0.0005), its error shrinks with q_c and vanishes at $q_c = 24$ (Figure 5). It is larger under probability matching because those swarms stay mixed, which also makes gate mistakes more common: the controller stays silent although the true share is below 0.75 on 5–8% of nights, and posts although it is above 0.75 on up to 13% at $q_c = 3$. Abstaining agents (0.1–0.2% of agent-days) do not bias it.

Helping the truth versus agents' own proofs: does not apply here. In task003-symmetric, extra truth posts raise the share of agents holding a proof of A0 but lower the share holding one from their own evidence. In task003-nosolution no proof is possible by construction, so this question does not arise.

6 How efficiently does the controller steer?

A controller can change the swarm in two ways: in the direction it is asked to, or in ways that happen whatever its target (for example pulling agents out of A1, or splitting them). We call the share of the first kind its *efficiency*. The target is chosen by a fair coin, so how much the swarm's state reveals about it is a well-defined information.

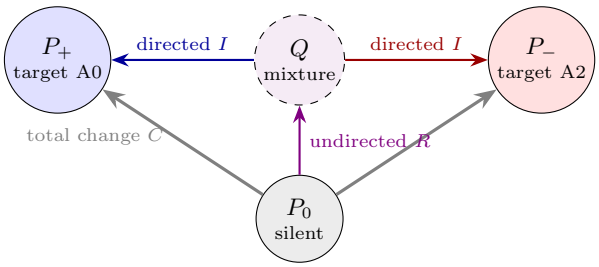


Diagram 2. Three distributions of the swarm's state at time t, one per arm, each over the episodes of that arm. The controller's total change C splits exactly into a part that depends on its target (I, from the mixture Q to each arm) and a part that does not (R, from silence to Q). Efficiency $\eta = I/C$ is the directed share.

For each arm a (silent 0, target A0 +, target A2 −) and a fixed time t, each episode e contributes a probability vector $y_a(e)$ over the values x of the state X, and

$$\hat{P}_a(x) = \frac{1}{n}\sum_{e=1}^{n} y_a(e)(x), \qquad I = \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_+\|\hat{Q}) + \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_-\|\hat{Q}), \qquad C = \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_+\|\hat{P}_0) + \tfrac{1}{2}\,\mathrm{KL}(\hat{P}_-\|\hat{P}_0), \quad (6)$$

with $\hat{Q} = \frac{1}{2}(\hat{P}_+ + \hat{P}_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$. I is the mutual information $I(Z;X_t)$ between the target Z (A0 or A2, chosen by a fair coin) and the state X_t; silence does not enter it. The reference $\hat{P}_0$ in C is the distribution of the *silent* arm. Exactly $C = I + \mathrm{KL}(\hat{Q}\|\hat{P}_0)$, and $\eta = I/C \in [0,1]$. Two resolutions: **one agent** (X = the vote of a randomly chosen agent; $y_a(e)$ = that episode's vote shares) and the **swarm** ($X = s = m_{\mathrm{A0}} - m_{\mathrm{A2}}$, the difference of the agents' mean beliefs, in three equal bins on $[-1,1]$; $y_a(e)$ marks the episode's bin). At swarm level the controlled swarm often reaches states the silent swarm never visits, which makes C infinite; there we use the bounded version $\eta_{JS} = I/(I + \mathrm{JS}(\hat{Q},\hat{P}_0))$, with JS the Jensen–Shannon divergence. We also measure how unsure the agents become: $\bar{H}$, the mean entropy of the agents' beliefs (0 to 1.58 bits), and the disagreement $\mathrm{JSD} = H(\text{mean belief}) - \bar{H}$, each as controlled minus silent in the same episode ($\Delta\bar{H}$, ΔJSD). All quantities are taken at one time, across episodes; they are not divergences between whole trajectories (those would be at least as large). Setting: probability matching, $q = 6$, $q_c = 12$ (Figure 7: every q_c); 1,000 episodes per arm.

What η does and does not say. $\eta = 1$ when the two controllers move the swarm by equal amounts in opposite directions around silence. It falls in two cases: both controllers push the swarm the same way (a *common push*), or one moves it much more than the other (a *lopsided* response, for example when silence is already close to one target). η measures how directed the change is, not how large it is (I) or what it costs; information per controller message (I divided by the messages sent) varies far more across settings.

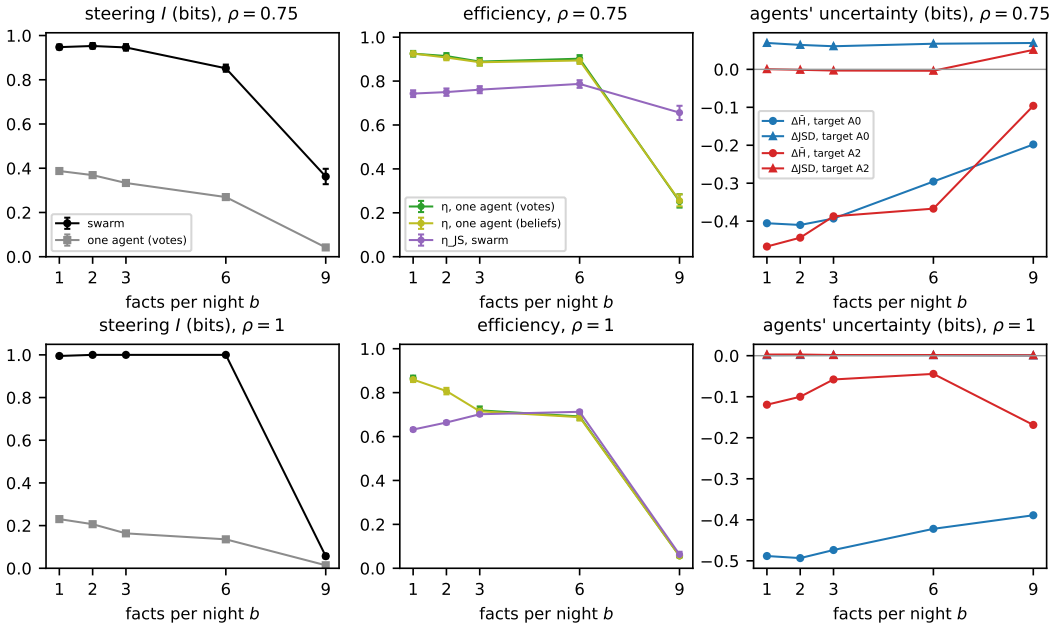


Figure 6: Steering I (left), efficiency (middle) and the change in the agents' uncertainty (right; circles $\Delta\bar{H}$, triangles ΔJSD) at $t = 30$ against facts per night b ($q = 6$, $q_c = 12$). Bars: 95% intervals.

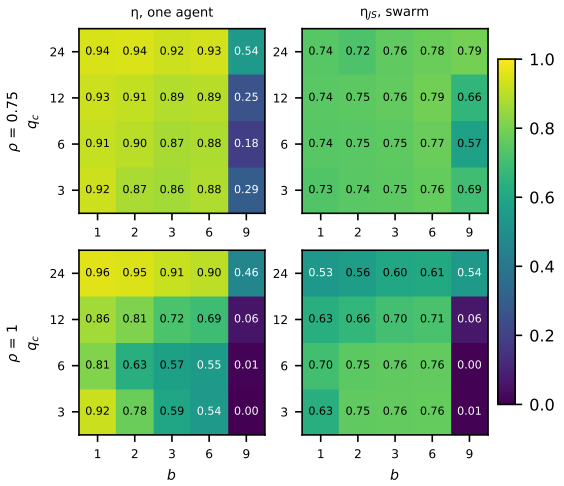


Figure 7: Phase diagram of efficiency at $t = 30$ over the controller's reading budget q_c and the facts per night b ($q = 6$): one-agent η (beliefs) and swarm η_{JS}. Dash: undefined (the controller changes almost nothing).

Table 4: Precision. Value from 1,000 paths per arm, and root-mean-square error of the estimate from 50, 100 or 300 paths (200 random subsets each), at $t = 30$, $b = 1$, $\rho = 0.75$. *: share of subsets in which the quantity is infinite or undefined.

quantity	1,000 paths	error, 50	error, 100	error, 300
steering I, swarm (3 bins)	0.948	0.031	0.022	0.010
steering I, one agent (votes)	0.387	0.027	0.018	0.010
efficiency η, one agent (votes)	0.925	0.030	0.021	0.010
efficiency η_{JS}, swarm	0.743	0.037	0.025	0.013
$\Delta\bar{H}$, target A0	-0.406	0.037	0.026	0.012
$\Delta\bar{H}$, target A2	-0.467	0.032	0.025	0.013
ΔJSD, target A0	0.070	0.010	0.007	0.003
ΔJSD, target A2	0.001	0.010	0.007	0.004

Findings. (Figure 6; probability matching.)

- **The large-budget backfire is a collapse of efficiency.** Up to $b = 6$ the change is almost all directed (one agent: $\eta = 0.89$–0.93 at $\rho = 0.75$); at $b = 9$ it drops to 0.25 ($\rho = 0.75$) and 0.06 ($\rho = 1$), and the steering I collapses with it. The extra facts move the swarm, but not in a target-dependent way.
- **With full memory, efficiency falls step by step with** b (0.87 at $b = 1$, 0.69 at $b = 6$; at $q_c = 24$ it stays above 0.9, Figure 7): the more of the pool goes out, the more both controllers push the swarm toward the same place (a common push toward A0).
- **Controllers make agents surer** ($\Delta\bar{H} \approx -0.4$ bits at $b \leq 3$, $\rho = 0.75$); the truth controller also splits them a little (ΔJSD +0.06 to +0.07). At $b = 9$ the gain in certainty shrinks.
- **The swarm-level** I **saturates near 1 bit** with full memory: every episode ends on its controller's side, however small the push. I measures how distinguishable the outcomes are, not how large the effect is; the gains of Section 4 measure size.
- **100 paths**: errors of about ±0.02–0.04 (Table 4).

Caveats. One world (task_003), 24 agents, exact-Bayesian agents; means over 1,000 episodes per cell. Full analysis, code and tables: `analysis/task003_llm_free/sim1_analysis/` (reports Q0–Q6).